

An Integral Segment with a Non-Real-Rooted Narayana-Weighted Lattice-Point Sum

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China

xiaojianzeng@foxmail.com

September 11, 2026

Abstract

For the Narayana polynomials

$$N_m(t) = \sum_{j=1}^m \frac{1}{m} \binom{m}{j} \binom{m}{j-1} t^j,$$

Defant asked whether there is an integral polytope in a positive equal-coordinate-sum hyperplane whose full, unit-weight lattice-point sum of products of the polynomials N_m is not real-rooted. We give the segment

$$P = \text{conv}\{(1, 4, 4), (3, 3, 3)\} \subset \{x \in \mathbb{R}_{>0}^3 : x_1 + x_2 + x_3 = 9\}.$$

Its only lattice points are its two endpoints, and its weighted sum is

$$t^3(2t^6 + 21t^5 + 78t^4 + 119t^3 + 78t^2 + 21t + 2).$$

An exact reciprocal-cubic reduction shows that the degree-six factor has two real zeros and four nonreal zeros, counted with multiplicity. A rational Sturm certificate gives an independent check. This resolves the stated existence question; no minimality, classification, or priority claim is made.

1 The question and the witness

For a positive integer vector $q = (q_1, \dots, q_r)$, write

$$N_q(t) = \prod_{i=1}^r N_{q_i}(t).$$

Defant's Question 6.2 asks, in particular, whether one can exhibit an integral polytope

$$P \subseteq \{x \in \mathbb{R}_{>0}^r : x_1 + \dots + x_r = L\}$$

for which the polynomial

$$F_P(t) = \sum_{q \in P \cap \mathbb{Z}^r} N_q(t)$$

is not real-rooted [1]. Here every lattice point of P occurs once; the example below has no hidden selection or mixing coefficients.

The general tactic of pairing a small explicit witness with exact root certificates also occurs in adjacent work on real-rootedness questions for combinatorially defined polynomials [2]. That work concerns preorder support polynomials, not Narayana-weighted lattice-point sums, and is cited only for this broad methodological parallel.

Theorem 1. *Let $r = 3$, $L = 9$, and*

$$P = \text{conv}\{(1, 4, 4), (3, 3, 3)\}.$$

Then

$$P \cap \mathbb{Z}^3 = \{(1, 4, 4), (3, 3, 3)\},$$

and F_P has exactly four nonreal zeros, counted with multiplicity. Consequently, P answers the existence part of Defant's Question 6.2.

Proof. Both endpoints are positive integral vectors with coordinate sum 9, so P is an integral polytope in the required hyperplane. Each point of P has the form

$$(1, 4, 4) + \lambda(2, -1, -1) = (1 + 2\lambda, 4 - \lambda, 4 - \lambda), \quad 0 \leq \lambda \leq 1.$$

If this point is integral, its second coordinate forces λ to be an integer. Thus $\lambda \in \{0, 1\}$, proving the displayed lattice-point list.

The definition gives

$$N_1(t) = t, \quad N_3(t) = t + 3t^2 + t^3, \quad N_4(t) = t + 6t^2 + 6t^3 + t^4.$$

Because the two endpoints are all the lattice points, each with weight one, direct integer convolution yields

$$\begin{aligned} F_P(t) &= N_1(t)N_4(t)^2 + N_3(t)^3 \\ &= t^3 G(t), \end{aligned}$$

where

$$G(t) = 2t^6 + 21t^5 + 78t^4 + 119t^3 + 78t^2 + 21t + 2. \quad (1)$$

It remains to count the nonreal zeros of G . Set

$$H(y) = 2y^3 + 21y^2 + 72y + 77.$$

For $t \neq 0$, the identities $t^2 + t^{-2} = y^2 - 2$ and $t^3 + t^{-3} = y^3 - 3y$, with $y = t + t^{-1}$, give the polynomial identity

$$G(t) = t^3 H(t + t^{-1}). \quad (2)$$

The exact cubic discriminant and the value at the endpoint of the relevant real interval are

$$\text{Disc}(H) = -1296 < 0, \quad H(-2) = 1.$$

Hence H has a unique real zero y_0 and a nonreal conjugate pair. Since $H(y) \rightarrow -\infty$ as $y \rightarrow -\infty$, the value $H(-2) = 1$ implies $y_0 < -2$.

Let y_0, y_1, y_2 be the three zeros of H . Equation (2) factors as

$$G(t) = 2 \prod_{i=0}^2 (t^2 - y_i t + 1).$$

The factor associated with $y_0 < -2$ has two distinct real zeros. Neither factor associated with a nonreal y_i can have a real zero: if a nonzero real t solved $t^2 - y_i t + 1 = 0$, then $y_i = t + t^{-1}$ would be real. Those two quadratic factors therefore contribute four nonreal zeros. The discriminant of H is nonzero, and none of its zeros is 2 or -2 , so these six zeros of G are simple. Finally, the factor t^3 in F_P only adds the real zero 0. Thus F_P has four nonreal zeros, counted with multiplicity. $\square$

2 A direct rational Sturm certificate

For completeness, non-real-rootedness can also be checked without passing to the reciprocal cubic. A signed Euclidean Sturm sequence for G , with each member multiplied by a positive rational number to make its integer coefficients primitive, is

$$\begin{aligned} S_0 &= 2t^6 + 21t^5 + 78t^4 + 119t^3 + 78t^2 + 21t + 2, \\ S_1 &= 4t^5 + 35t^4 + 104t^3 + 119t^2 + 52t + 7, \\ S_2 &= 37t^4 + 252t^3 + 417t^2 + 224t + 33, \\ S_3 &= -521t^3 - 630t^2 - 126t - 7, \\ S_4 &= -10405t^2 - 11459t - 1997, \\ S_5 &= -3581t - 378, \\ S_6 &= 1. \end{aligned}$$

At $-\infty$ their signs are

$$(+, -, +, +, -, +, +),$$

which have four variations. At $+\infty$ their signs are

$$(+, +, +, -, -, -, +),$$

which have two variations. Sturm's theorem therefore gives exactly two distinct real zeros of G . The nonzero terminal constant also gives $\gcd(G, G') = 1$, so G is square-free. Since $\deg G = 6$, precisely four of its zeros are nonreal, with multiplicity.

3 Scope and reproducibility

The theorem is only an existence result. It does not assert that the witness is smallest under any ordering, characterize all eligible polytopes, or settle real-rootedness for special subclasses such as integral binary nestohedra.

The witness was found by a deterministic search over primitive integral segments. The accompanying Python program uses only standard-library integer and rational arithmetic. It independently enumerates all 28 positive compositions of 9 into three parts, checks exact segment membership, reconstructs the coefficient list of F_P , computes the square-free decomposition and Sturm chain, and verifies the reciprocal identity and cubic discriminant. The program and its expected JSON certificate are supplied as supplementary files. The search ordering is not used to support a minimality claim.

OpenAI Codex assisted with the bounded witness search, exact-certificate workflow, and manuscript preparation. The proof above is self-contained; the computational supplement makes the finite searches and algebraic cross-checks reproducible, but no numerical approximation or software verdict is used as a proof of non-real-rootedness.

References

- [1] C. Defant, *Fertilitopes*, arXiv:2102.11836v2 [math.CO], 2023. <https://arxiv.org/abs/2102.11836>.
- [2] Z. Zeng, *An Eight-Element Counterexample to Real-Rootedness of Preorder Support Polynomials*, 2026, DOI: 10.2139/ssrn.7385158.